THE UNIVERSITY OF ILLINOIS AT CHICAGO
Department of Chemical Engineering

AWARD TEAM HOMEWORK SET #6

Course: CHE 301. CHEMICAL ENGINEERING THERMODYNAMICS
Term: Fall 2017
Professor: Dr. Michael Caracotsios
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Your name: Vrushant Patel
Your name: Harsik Patel
Your name: ____________________
Due Date: Wednesday Nov-08, 2017 by 3:30pm

Number of points..............................100
Your score.................................... 94

Award: **Gift card to each member of the winning team**

To be considered for the award, the team homework final report must be typed. Homework results should be presented in a logical and concise manner, with all steps in derivation processes [if any] shown very clearly. The VBA code must be well documented and attached to the homework report. Images of the pertinent Excel worksheets must be clearly shown in the homework solution report. You may also want to include anything else (like external references) you believe will increase your chances of winning the award.

Remember: All members of the team that wins a total of three(3) awards will be given **100** points towards the final examination and will be excused from taking it.

Homework #6: Fugacity of liquid water and calculations using Raoult's law

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Problem (1)

Calculate the fugacity of liquid water at T = 250 degree Celsius and P = 70 bar using the Peng-Robinson equation of state.

For Peng-Robinson Equation of state

$$P + \frac{a\sigma(T_r)}{V(V+b)+b(V-b)} = \frac{RT}{V-b}$$

$$a = 0.457235\frac{(RT_c)^2}{P_c} \quad and \quad A = \frac{aP\sigma(T_r)}{(RT)^2}$$

$$b = 0.077796\frac{RT_C}{P_c} \quad and \quad B = \frac{bP}{RT}$$

$$\sigma(T_r) = [1 + \kappa\,(1 - \sqrt{T_r}\,)]^2 \text{ where } \kappa = 0.37464 + 1.54226\omega - 0.26992\omega^2$$

$$\boxed{\boldsymbol{Z = \frac{1}{(1-b\rho)} - \frac{a\sigma(T_r)}{RT}\left(\frac{\rho}{(1+b\rho)+b\rho(1-b\rho)}\right)}}$$

$$\boxed{\boldsymbol{\ln\phi = (Z-1) - \ln(Z-B) - \frac{A}{2\sqrt{2}B} * \left(\ln\left(\frac{Z+(1+\sqrt{2})B}{Z+(1-\sqrt{2})B}\right)\right)}}$$

1) Calculation of fugacity of liquid water using steam table.

Given that T = 250 Celsius and P = 70 bar.

Step 1: Determine the saturated pressure both using steam tables at temperature T = 250 degree Celsius.

For the Steam table, consider using Athena Visual Software to obtain data from the steam properties.

Please Turn Over

Gibbs free energy equation: $G = H - TS$

- G at P = P^{sat}

$$G = 2798.562\frac{kJ}{kg} - (259 + 273.15)K * 6.607\frac{kJ}{kg\ K}$$

$$\boxed{G = -6760.449\frac{kJ}{kg}}$$

- G at P = 1 bar

$$G^{IG} = 2972.524\frac{kJ}{kg} - (250 + 273.15)K * 8.083\frac{kJ}{kg\ K}$$

$$\boxed{G^{IG} = -22626.502\frac{kJ}{kg}}$$

Substituting the values into the equation to calculate the fugacity

$$\frac{G - G^{IG}}{RT} = ln\left(\frac{f}{P}\right)$$

$$\frac{-6760.449 - (-22626.502)\frac{kJ}{kg} * \frac{1000\,J}{1kJ} * \frac{1kg}{1000\,g} * \frac{18.015g}{1mol}}{8.314\frac{J}{mol\ K} * (250 + 273.15)\ K} = ln\left(\frac{f}{1\ bar}\right)$$

$$f = 1bar * e^{3.648} = 38.391\ bar$$

NOTE: The result is what is expected i.e. f < P, hence the given conditions are non-ideal and that water is real at given conditions.

Therefore, fugacity coefficient of the liquid water at saturated vapor pressure P^{sat} = 39.776 bar is

$$\boxed{f = 38.391\ bar}$$

Step 3: Calculating fugacity of liquid water at P = 70 bar and T = 250 degree Celsius by choosing P = P^{sat} and T = 250 degree as reference state, because at that state we know the fugacity of liquid water.

Data from Athena Visual Software for P = 70 bar and T = 250 degree Celsius

Therefore, fugacity coefficient of the liquid water at P = 70 bar is

$$\boxed{\phi = \frac{f}{P} = \frac{38.995\ bar}{70\ bar} = 0.5571} \quad ✓ \quad +8$$

2) The Peng-Robinson equation of state for both vapor and liquid phases.

Step 1: Compute the saturated vapor pressure using the Antoine equation by providing the temperature as the parameter.

Saturated vapor pressure from the Antoine equation = 37.079 bar

Step 2: Compute the saturated vapor pressure using the equation of state (EOS), by using the saturated vapor pressure from the Antoine equation as the initial guess.

Saturated vapor pressure from EOS = 40.119 bar

Step 3: Verify that the fugacity coefficient of the vapor and liquid phase is equal at the saturated vapor pressure computed using EOS, which confirms that equation of state holds good for water.

$$\phi_{sat}^{vapor} = \phi_{sat}^{liquid} = 0.8689$$

Step 4: Compute the fugacity coefficient of the liquid phase by providing the parameters of temperature in K, Pressure in bar, and initial guess for compressibility factor (Z) which is 0.

$$\phi^{liquid} = 0.5076$$ ✓ +8

3) The Peng-Robinson equation of state for the vapor phase and Poynting correction theory for the liquid phase.

Step 1: Compute the fugacity coefficient of the vapor phase by providing the parameters of temperature in K, Pressure in bar, and initial guess for compressibility factor (Z) which is 1.

$$\phi^{vapor} = 0.7716$$

Consider, now calculating fugacity coefficient of liquid water using Poynting correction theory using the equation below

$$\phi^{liquid} = \phi_{sat}^{vapor}\left(\frac{P^{sat}}{P}\right) exp\left(\frac{V^l}{RT} * (P - P^{sat})\right)$$ ✓

The colored term is called the Poynting correction factor

+16

VLE Calculations

Peng-Robinson

Fluid Conditions	Units	Using Steam Table	Using PR	Using Poynting correction factor						
Molar Composition		Part (1)	Part (2)	Part (3)						
Temperature	C	250.000	250.000	250.000						
Pressure	bar	70.000	70.000	70.000			FOR VAPOR			
							at Psat and 250 C			
Saturated Vapor Pressure: Antoine	bar		37.079	37.079			H (kJ/kg)	2798.582		
Saturated Vapor Pressure: EOS	bar		40.119	40.119			S (kJ/kg K)	6.067		
Saturated Vapor Pressure: Steam Table	bar	39.776					G (kJ/mol)	-6760.449	Fugacity vapor/liquid	38.391
									at Psat	
Saturated Temperature: Antoine	C						at 1 bar and 250 C			
Saturated Temperature: EOS	C						H	2972.524	G (kJ/kg)	-22626.502
							S	6.083		
Vapor Phase	**Units**									
Compressibility Factor					PHIC sat vapor	PHIC sat liquid	FOR LIQUID	at 70 bar and 250 C		
Fugacity Coefficient				0.7716	0.8689	0.8689	H (kJ/kg)	1085.028		
							S (kJ/kg K)	2.784		
Molar Density	kmol/m3						G (kJ/kgl)	-6694.485	Fugacity of liquid	38.995
Molar Enthalpy	kJ/kmol								at 70 bar	
Molar Entropy	kJ/(kmol K)									
									Fugacity coeffecient	0.5571
Liquid Phase	**Units**									
Compressibility Factor							FOR LIQUID			
Fugacity Coefficient		0.5571	0.5076	0.5064	Zc	0.2290	at Psat and 250 C			
Poynting factor				1.017	Tr	0.808	H (kJ/kg)	1085.058		
Molar Density	kmol/m3			40.706	Pc	217.600	S (kJ/kg K)	2.792		
Molar Enthalpy	kJ/kmol				Tc	374.150	G (kJ/kg)	-6782.420		
Molar Entropy	kJ/(kmol K)				Vc	55.895				
					Liquid Volume	2.457E-05				
					Rackett Factor	0.243172				

VLE Calculations							Ideal Gas
Fluid Conditions	**Units**	**C5H12**	**C7H16**	**C10H22**	**Mixture**		
Molar Composition		0.816700	0.164600	0.018700	1.000000		
Temperature	C				42.373		
Pressure	bar				1.01325		
							Sum(Zi * Pi sat) - P
Saturated Vapor Pressure: Antoine	bar	1.214	0.134	0.006			0.000
Saturated Vapor Pressure: EOS	bar						
Saturated Temperature: Antoine	C						
Saturated Temperature: EOS	C						
K-Values							
Vapor to Feed Molar Ratio							
Rachford-Rice Equation							
Dew Point Pressure	bar						
Bubble Point Pressure	bar						
Dew Point Temperature	C						
Bubble Point Temperature	C	42.373					

Step 4: For dew point temperature, we know that $\alpha = 1$. Thus, we get an equation from the Rachford-Rice equation for dew point temperature/pressure.

$$\sum_{i=1}^{N} \frac{z_i(k_i - 1)}{1 + \alpha(k_i - 1)} = 0$$

$$\sum_{i=1}^{N} \frac{z_i(k_i - 1)}{k_i} = 0$$

We can simplify the bracket, to get following equation.

$$\sum_{i=1}^{N} \frac{z_i}{k_i} * k_i - \sum_{i=1}^{N} \frac{z_i}{k_i} = 0$$

The k_i from the first term cancels out to result in following expression.

$$\sum_{i=1}^{N} z_i - \sum_{i=1}^{N} \frac{z_i}{k_i} = 0$$

The highlighted term in the above expression is 1, and we replace k_i by $\frac{P_i^{sat}}{P}$

$$\sum_{i=1}^{N} \frac{z_i}{P_i^{sat}} * P = 1$$

$$P^{DEW} = \frac{1}{\sum_{i=1}^{N} \frac{z_i}{P_i^{sat}}}$$

Step 1: We know that the temperature should be somewhere in between bubble and dew point temperature, so that vapor-liquid mixture is formed which is eventually separated.

Step 2: We made a guess for the temperature to be 60 which is in-between bubble and dew point temperature and found that Rachford-Rice equation is almost satisfied and composition of n-decane is close to the target.

Step 3: We tried to add decimals after 60 (since we were close) to get a value for the temperature.

Finally, the temperature of condenser that would yield liquid distillate molar fraction of n-decane equal to 0.137 is **60.92 degree Celsius** and the alpha value for which Rachford-Rice equation is satisfied is **0.8773**. +20

The results for the problem can be seen below.

VLE Calculations					Ideal Gas			
Fluid Conditions	Units	C5H12	C7H16	C10H22	Condensor			
Molar Composition		0.816700	0.164600	0.018700	1.000000			
Temperature	C				60.920			
Pressure	bar				1.01325			
Saturated Vapor Pressure: Antoine	bar	2.120	0.284	0.016				
Saturated Vapor Pressure: EOS	bar							
Saturated Temperature: Antoine	C							
Saturated Temperature: EOS	C							
							Temperature of the Condensor	60.920
K-Values		2.0926	0.2808	0.0158				
Vapor to Feed Molar Ratio					0.8773			
Rachford-Rice Equation					0.0000		Alpha	0.8773
Dew Point Pressure	bar							
Bubble Point Pressure	bar							
Dew Point Temperature	C	80.129						
Bubble Point Temperature	C	42.373						
Vapor Phase	Units							
Molar Composition		0.8726	0.1252	0.0022	1.0000			
Compressibility Factor								
Fugacity Coefficient								
Molar Density	kmol/m3							
Molar Enthalpy	kJ/kmol							
Molar Enthalpy of Vaporization @Tb	kJ/kmol							
Molar Entropy	kJ/(kmol K)							
Molar Entropy of Vaporization @Tb	kJ/(kmol K)							
K-values								
Liquid Phase	Units							
Molar Composition		0.4173	0.4461	0.1369	1.0000			
Compressibility Factor								
Fugacity Coefficient								

3) Calculate the duty of the overhead condenser for the temperature in part (2).

Approach

The way to solve the problem is to write an energy balance equation for the feed stream ($V_1 = V_F$) going into the condenser and the vapor and liquid stream leaving the separator.

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Step 1: Calculating the enthalpy of each component in feed stream at temperature 80.129 degree Celsius using the Hmig formula in the CHE_Toolbox for the ideal gas.

Step 2: Calculate the total enthalpy of feed stream by using the molar composition of the feed stream.

$$H_F = -147464.059\ kJ/kmol$$

Step 3: Calculating the enthalpy of each component in vapor phase at temperature 60.92 degree Celsius using the Hmig formula in the CHE_Toolbox for the ideal gas.

Step 4: Calculate the total enthalpy of vapor phase by using the molar composition of the vapor phase.

$$H^V = -147082.873\ kJ/kmol$$

Step 5: On a separate worksheet incorporate the critical temperature, boiling temperature, and heat of evaporation at boiling temperature (cal/mol) for each component.

Make a column for temperature with various values at appropriate interval and heat of vaporization which will be calculated for that temperature.

By writing the formula mentioned in the approach using the cell reference in the Excel we calculate the Heat of vaporization at desired temperature of 60.92 degree Celsius.

Below are the results

	n-Pentane
Critical Temperature, C	196.45
Normal Boiling Point, C	36.05
Heat of Vaporization, cal/mol	6160
	n-Heptane
Critical Temperature, C	267.05
Normal Boiling Point, C	98.45
Heat of Vaporization, cal/mol	7576
	n-Decane
Critical Temperature, C	344.45
Normal Boiling Point, C	174.15
Heat of Vaporization, cal/mol	9388

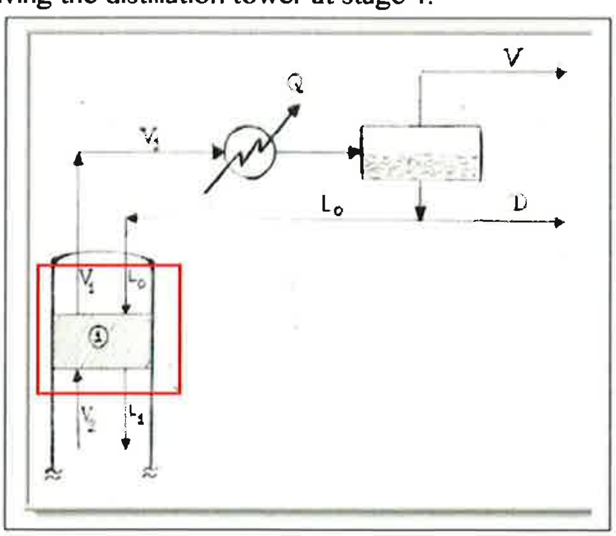
n-Pentane		n-Heptane		n-Decane	
T	DHvap	T	Dhvap	T	Dhvap
0.00	6646.59	0.00	9002.03	0.00	12226.16
2.00	6621.13	2.00	8976.69	2.00	12199.49
4.00	6595.51	4.00	8951.23	4.00	12172.72
6.00	6569.72	6.00	8925.65	6.00	12145.85
8.00	6543.76	8.00	8899.94	8.00	12118.89
10.00	6517.63	10.00	8874.11	10.00	12091.82
12.00	6491.33	12.00	8848.16	12.00	12064.66
14.00	6464.84	14.00	8822.08	14.00	12037.39
16.00	6438.18	16.00	8795.86	16.00	12010.02
18.00	6411.32	18.00	8769.52	18.00	11982.54
20.00	6384.28	20.00	8743.04	20.00	11954.96
22.00	6357.05	22.00	8716.43	22.00	11927.27
24.00	6329.62	24.00	8689.69	24.00	11899.47
26.00	6301.99	26.00	8662.81	26.00	11871.57
28.00	6274.16	28.00	8635.78	28.00	11843.55
30.00	6246.12	30.00	8608.62	30.00	11815.43
32.00	6217.87	32.00	8581.31	32.00	11787.19
34.00	6189.41	34.00	8553.85	34.00	11758.84
36.00	6160.72	36.00	8526.25	36.00	11730.37
38.00	6131.81	38.00	8498.50	38.00	11701.79
40.00	6102.67	40.00	8470.60	40.00	11673.10
42.00	6073.30	42.00	8442.54	42.00	11644.28
44.00	6043.69	44.00	8414.32	44.00	11615.35
46.00	6013.83	46.00	8385.95	46.00	11586.29
48.00	5983.73	48.00	8357.42	48.00	11557.11
50.00	5953.37	50.00	8328.72	50.00	11527.81
52.00	5922.75	52.00	8299.86	52.00	11498.39
54.00	5891.86	54.00	8270.83	54.00	11468.84
56.00	5860.70	56.00	8241.63	56.00	11439.16
58.00	5829.27	58.00	8212.25	58.00	11409.35
60.00	5797.55	60.00	8182.70	60.00	11379.41
60.92	5782.86	60.92	8169.06	60.92	11366.60
62.00	5765.53	62.00	8152.97	62.00	11349.34
64.00	5733.22	67.00	8077.84	72.00	11196.96
66.00	5700.60	72.00	8001.53	77.00	11119.46
68.00	5667.67	77.00	7923.99	82.00	11041.04
70.00	5634.41	82.00	7845.16	87.00	10961.69

Fluid Conditions	Units	C5H12	C7H16	C10H22	Feed			
Molar Composition		0.816700	0.164600	0.018700	1.000000			
Temperature	C				80.129			
Pressure	bar				1.01325			
Saturated Vapor Pressure: Antoine	bar	3.510	0.558	0.041				
Saturated Vapor Pressure: EOS	bar							
Molar Enthalpy of the feed	kJ/kmol	-138304.729	-177820.145	-235734.272	-147464.059			
Saturated Temperature: EOS	C							
							Temperature of the Condenser	80.920
K-Values		3.4640	0.5515	0.0402				
Vapor to Feed Molar Ratio								
Rachford-Rice Equation							Alpha	0.8773
Dew Point Pressure	bar							
Bubble Point Pressure	bar						Duty of condesor (Q) (kJ/kmol)	8.721
Dew Point Temperature	C	80.128						
Bubble Point Temperature	C	42.373						
Vapor Phase	**Units**							
Molar Composition		0.8726	0.1252	0.0022	1.0000			
Compressibility Factor								
Fugacity Coefficient								
Molar Density	kmol/m3							
Molar Enthalpy	kJ/kmol	-141806.423	-181513.947	-240813.397	-147082.873			
Molar Enthalpy of Vaporization @60.92	kJ/kmol	24195.486	34179.305	47553.670				
Molar Entropy	kJ/(kmol K)							
Molar Entropy of Vaporization @Tb	kJ/(kmol K)							
Temperature of Condensor	C							
Liquid Phase	**Units**							
Molar Composition		0.4170	0.4461	0.1369	1.0000			
Compressibility Factor								
Fugacity Coefficient								
Molar Density	kmol/m3							
Molar Enthalpy	kJ/kmol	-166101.939	-215693.246	-288364.068	-204962.294			
Molar Entropy	kJ/(kmol K)							

3) Calculate the molar flow rate of liquid stream L_1 that is leaving the top stage of distillation tower and molar flow rate of vapor stream V_2 that is entering the distillation tower if the reflux ratio is 1.25. Vapor stream V_2 is at its dew point temperature, assume an initial guess for this temperature to be T_2 = 85.0 degree Celsius.

Approach

The way to solve the problem is write mass, energy, and component balance equations for the streams entering and leaving the distillation tower at stage 1.



The molar composition of x_1 and x_2 in L_0 is 0.4170 and 0.4461 respectively.

Step 2: Find the molar composition of x_1 and x_2 for stream L_1, by using the fact that it in equilibrium with stream V_1.

$$x_1 = \frac{y_1}{k_1} = \frac{0.8167}{3.464} = 0.2358$$

$$x_2 = \frac{y_2}{k_2} = \frac{0.1646}{0.5515} = 0.2985$$

You had correct eqn's, were you able to calculate L_1 and V_2?

+20

References

- Lecture Block #5 – Professor Michael Caracotsios Notes

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